



Golden Jubilee Lecture

Randomised Clinical Trials of Choroidal Melanoma Treatment

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Purpose: To illustrate an approach to evidence-based medical practice by reporting the Collaborative Ocular Melanoma Study (COMS) randomised clinical trials and cohort studies of choroidal melanoma.

Methods: COMS randomised clinical trials of Iodine-125 (I-125) brachytherapy, adjunctive cohort study of visual acuity in eyes treated with brachytherapy and adjunctive natural history study. COMS randomised clinical trial of pre-enucleation radiation.

Results: The COMS I-125 brachytherapy trial (N = 1,317 patients) of medium-sized choroidal melanoma showed 5-year all-cause mortality of 18% [95% Confidence Interval (CI), 16-20%] and no statistically significant difference in mortality following I-125 brachytherapy or enucleation. Adjunctive cohort natural history study (N=42 patients) of patients eligible for the I-125 brachytherapy trial who deferred treatment or had no melanoma treatment had a 5-year all-cause mortality of 30% (95% CI, 18-47%). The COMS pre-enucleation radiation trial (N = 1,003 patients) of large-sized choroidal melanoma showed 5-year all-cause mortality of 40% (95% CI, 37-44%).

Conclusions: Evidence derived from randomised clinical trials and cohort studies shows the need for longterm (≥ 5 years) follow-up to determine the efficacy of treatment for choroidal melanoma by any modality. The rather similar 5-year mortality for treated and untreated medium melanoma patients suggests that metastatic dissemination may occur at an early stage of choroidal melanoma. To increase longterm survival, ocular treatment of choroidal melanoma must strive for diagnosis and treatment of melanoma at an early stage when metastasis is less likely and be combined with measures to detect and treat micrometastasis.

Key Words: Melanoma, choroid, evidence, clinical trial, brachytherapy, cohort study.

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Celebrating 50 years of publication as the official journal of the All India Ophthalmological Society, the *Indian Journal of Ophthalmology* warrants all commendation and applause. For half a century, the *Indian Journal of Ophthalmology* has received the clinical observations, creative thoughts and original ideas of ophthalmic scientists, nurtured these contributions through scholarly peer review, and archived these scientific products through publication and worldwide distribution.

Led by a succession of distinguished editors (Table 1), the Indian Journal of Ophthalmology has particular relevance to the populations of India, the geographic diversity of this large region and the clinical challenge

presented to ophthalmologists by a population of more than one billion people. More broadly, the *Journal* is an important part of the mosaic of scientific journals published throughout the globe.

To translate the avalanche of new information from clinical research to clinical practice, ophthalmologists need to apply the principles of evidence-based medicine and ask three questions: (1) Are the results of the study valid? (2) What are the results? (3) Do these results help me in caring for my patients?¹⁻⁵

To answer these questions, it is important to understand the strengths and limitations of scientific study design and methodology. Case reports, for example, and consecutive case series may provide important insights regarding disease processes, describe new surgical procedures or generate hypotheses. Case series, however, describe clinical experiences and, with rare exceptions, do not produce scientific evidence. Case-control studies select people on the basis of the disease presence (cases) or disease absence (controls) and may be practice-based or population-based. Case-control studies are useful for rare diseases and may identify hypotheses, but cannot measure relative risk or disease incidence.⁵

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Table 1. Editors of the Indian Journal of Ophthalmology, 1953-2003

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Providing evidence, cohort studies are prospective, longitudinal studies that select people on the basis of disease presence or disease absence, measure characteristics at the onset and follow participants periodically to monitor outcome. Because these studies are prospective and carried out over a period of time, cohort studies measure the incidence of disease or other outcome as well as relative risk. One type of cohort study is the natural history study in which people with the disease are followed to determine longterm outcome and risk factors.

Providing the highest level of experimental evidence, randomised clinical trials generally select people with specific disease characteristics, randomly assign the participants to various treatment options, minimise bias by masking the observers, and measure outcome. A clinical trial may be appropriate when (1) the disease to be studied is a substantial public health concern, (2) preliminary evidence exists that a “new” test treatment may be superior to the “standard” treatment, (3) a valid trial outcome may be obtained in a reasonable time, and (4) ethical patient safeguards are in place.^{6,7}

The Collaborative Ocular Melanoma Study (COMS; Table 2), in progress since 1985 at 43 clinical centers in the United States and Canada illustrates use of the randomised clinical trial, cohort study and case series to build an evidence-based programme of choroidal melanoma management.

Table 2. Collaborative Ocular Melanoma Study

Chair	Stuart L. Fine, MD
Vice Chairs	John D. Earle, MD Bradley R. Straatsma, MD
Coordinating Centre Director	Barbara S. Hawkins, PhD
National Eye Institute Representative	Natalie Kurinij, PhD
Conducted at 43 Clinical Centers in the United States and Canada	
Sponsored by the National Eye Institute and the National Cancer Institute, United States	

The COMS randomised trial of Iodine-125 (I-125) brachytherapy for choroidal melanoma screened 8,712 choroidal melanoma patients; 5,046 had melanoma of eligible size; 2,882 fulfilled all eligibility criteria, and 1,317 enrolled in the trial.⁸⁻¹⁰ Enrolled patients had medium-sized choroidal melanoma (2.5 mm to 10 mm in apical height, no more than 16 mm in longest basal diameter, and with exceptions, proximal border 2 mm or more from the optic disc border). Eligible patients were adults, free of detectable melanoma metastasis and with no history of other primary cancer.⁸

Approximately half (1,371; 46% of 2,882) of the eligible patients were enrolled. Baseline demographic characteristics and tumour characteristics of enrolled and not-enrolled patients were well balanced for all factors important in determining outcome. Consequently, results of enrolled patients may be generalised to include eligible but not-enrolled patients.

To determine whether the enrolled patients are broadly representative of patients who would be candidates for treatment with I-125 brachytherapy, it was estimated that the 8,718 patients screened at COMS centers comprised 34-45% of all new cases of choroidal melanoma that occurred in the United States and Canada during the 11.5 years of COMS patient accrual. Based on the high proportion of melanoma cases screened at COMS Centers, it is reasonable to conclude that results of the COMS randomised trial of I-125 brachytherapy are generalisable to future patients who meet COMS eligibility criteria.

In the randomised trial of I-125 brachytherapy, 1,317 patients were enrolled; 660 were assigned randomly to enucleation, and 657 to I-125 brachytherapy.⁹ When histopathology was reviewed centrally, 658 (99.7%) of 660 eyes had choroidal melanoma and two eyes had carcinoma metastatic to the choroid. The I-125 brachytherapy protocol prescribed a radiation dose of 85 Gray (GY) at the tumour apex. The protocol utilised standard gold plaques that provided a 2-3 mm margin of treatment beyond the tumour border (Figure 1).

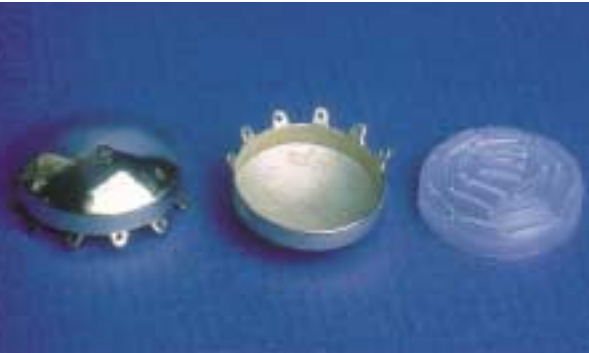


Figure 1. Standardised gold plaque with eyelets for fixation sutures and silastic insert for I-125 seeds.



For a typical case (Figure 2), the I-125 plaque placement procedure was performed under monitored local anaesthesia or general anaesthesia. A conjunctival peritomy was performed, sutures were looped around the rectus muscle insertions, the melanoma was localised by transillumination with ophthalmoscopy, and the sclera was marked to position the central meridian of the melanoma and identify the placement of each fixation suture. With the plaque in place and slipknots on fixation sutures, plaque placement was confirmed by intraoperative ultrasonography to ensure a 3 mm margin of treatment beyond the tumour at 90° intervals around the circumference of the melanoma (Figure 3). After confirmation of precise plaque placement, fixation sutures were tied and conjunctiva was repositioned to the limbus.

Based on date of enrollment, 1,072 (81%) of the patients had been followed after treatment for 5 years and 416 (32%) for 10 years. The unadjusted estimated 5-year mortality rates were 19% for enucleation and 18% for brachytherapy; there was no statistically significant difference in mortality rates ($P = 0.48$).⁹ For 1,317 patients in the I-125 brachytherapy trial, the Kaplan-Meier estimate of 5-year all-cause mortality was approximately 18% [95% Confidence Interval (CI), 16-20%].⁹

As an adjunct to the I-125 brachytherapy trial, a prospective non-randomised multicentre cohort study evaluated visual acuity in 623 eyes treated with I-125 brachytherapy and followed for at least one year.¹⁰ At baseline, before treatment, median visual acuity in the eye with melanoma was 6/9; 70% of eyes had 6/12 or better, and 10% of eyes had 6/60 or worse. Three years

Table 3. Iodine-125 Brachytherapy Trial: Visual Acuity at Baseline and three years after brachytherapy (n= 623 eyes)

Visual acuity*	Baseline	Three years after brachytherapy
Median	6/9	6/36
6/12 or better	70%	34%
6/60 or worse	10%	45%

* Best-corrected visual acuity with protocol refraction

after I-125 brachytherapy, median visual acuity was 6/36; 34% of eyes had 6/12 or better and 45% of eyes had 6/60 or worse; the latter included eyes that were enucleated (Table 3). Life-table estimates of patients who lost six or more lines of visual acuity (a quadrupling of the minimum angle of resolution) from baseline were 18% by one year, 34% by two years, and 49% by three years after treatment. Overall, about half of treated eyes had substantial impairment of visual acuity three years after I-125 brachytherapy, defined either by loss of six or more lines of visual acuity (49% of eyes) or 6/60 or worse best-corrected visual acuity (45% of eyes).¹⁰

Impairment of visual acuity after brachytherapy is shown in an eye with juxtafoveal melanoma and baseline visual acuity of 6/30 (Figure 4a). Two years following brachytherapy, the tumour had decreased in thickness, small retinal and choroidal blood vessels were obliterated in the area corresponding to the plaque and visual acuity was 6/120 (Figure 4b).

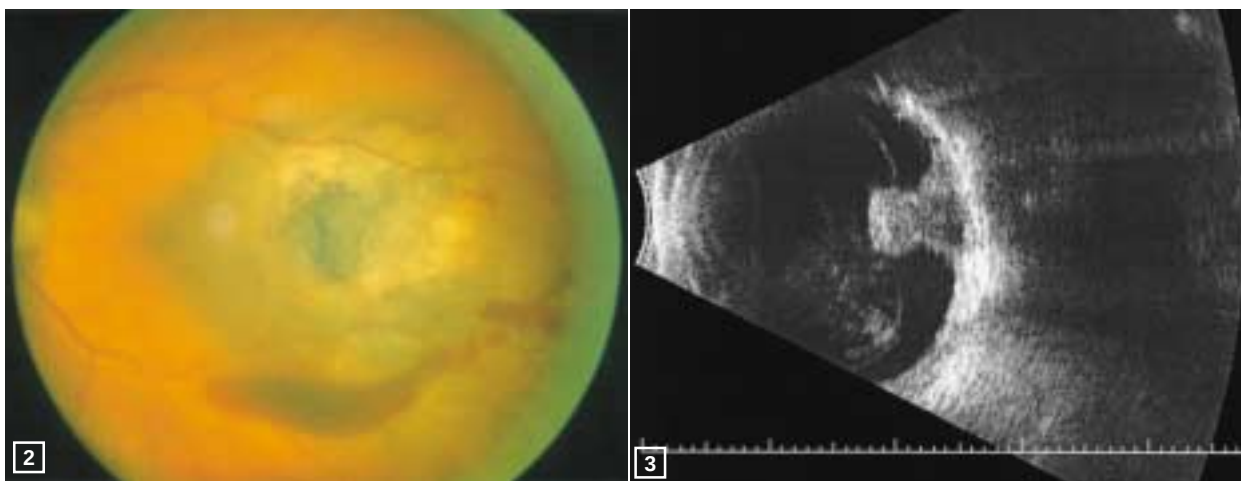


Figure 2. A 66-year-old man with medium-sized choroidal melanoma in the macula, left eye. Basal diameters of 9 mm and 11 mm and apical thickness of 5.8 mm prior to brachytherapy. Visual acuity: 6/60; **Figure 3.** Intraoperative ultrasonography confirming that the I-125 plaque extends 3 mm beyond the border of the melanoma. Tests confirm 3 mm margin at 90° intervals around the circumference. Cursors identify the tumour border and plaque edge. 85-year-old man with mushroom-shaped melanoma with basal diameters of 10 mm and 10 mm and apical thickness of 6.3 mm.



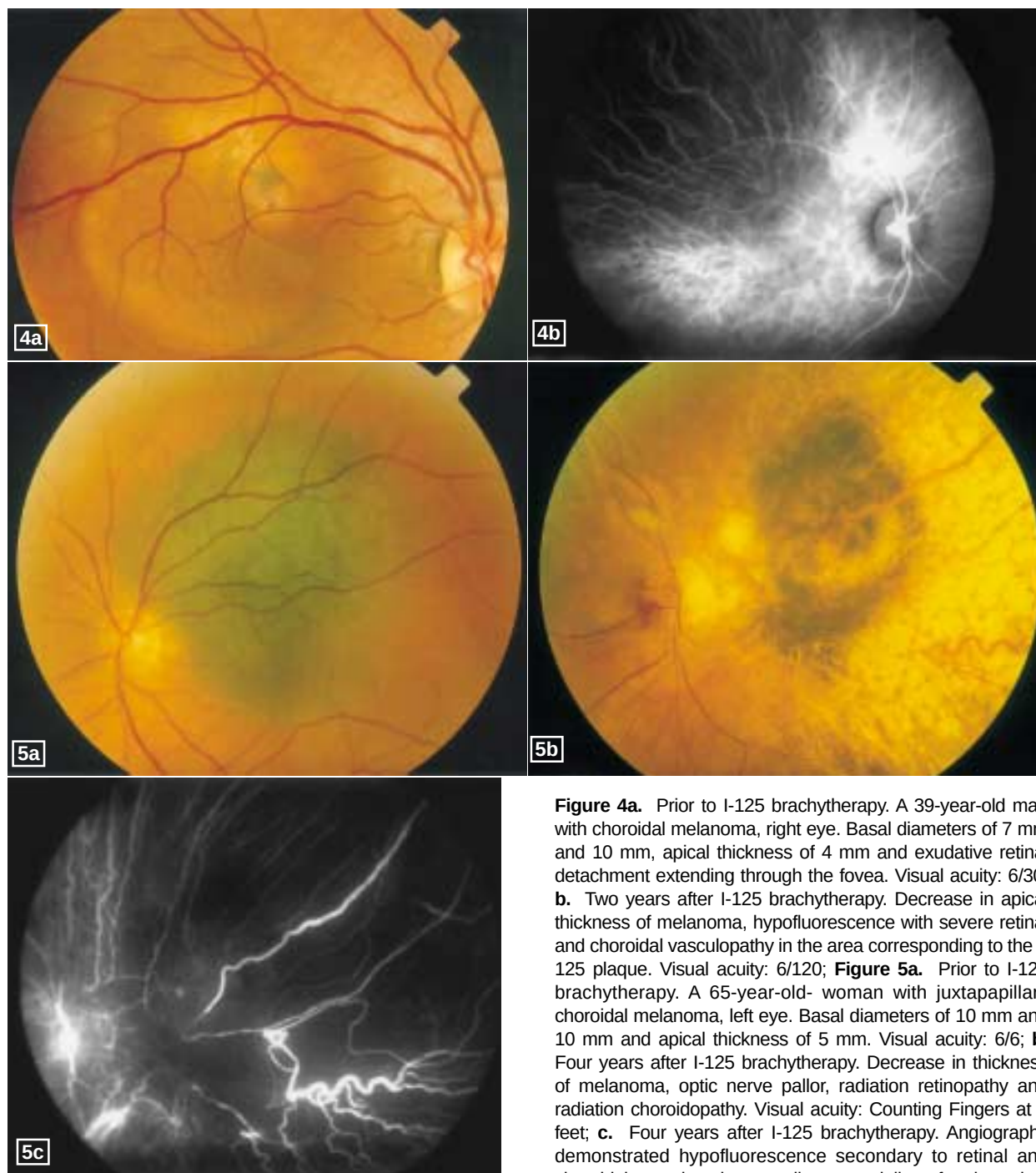


Figure 4a. Prior to I-125 brachytherapy. A 39-year-old man with choroidal melanoma, right eye. Basal diameters of 7 mm and 10 mm, apical thickness of 4 mm and exudative retinal detachment extending through the fovea. Visual acuity: 6/30; **b.** Two years after I-125 brachytherapy. Decrease in apical thickness of melanoma, hypofluorescence with severe retinal and choroidal vasculopathy in the area corresponding to the I-125 plaque. Visual acuity: 6/120; **Figure 5a.** Prior to I-125 brachytherapy. A 65-year-old woman with juxtapapillary choroidal melanoma, left eye. Basal diameters of 10 mm and 10 mm and apical thickness of 5 mm. Visual acuity: 6/6; **b.** Four years after I-125 brachytherapy. Decrease in thickness of melanoma, optic nerve pallor, radiation retinopathy and radiation choroidopathy. Visual acuity: Counting Fingers at 3 feet; **c.** Four years after I-125 brachytherapy. Angiography demonstrated hypofluorescence secondary to retinal and choroidal vasculopathy as well as remodeling of major retinal and choroidal vessels (Indocyanine green).

More extensive postradiation vascular alterations appear in an eye with juxtapapillary choroidal melanoma and visual acuity of 6/6 prior to brachytherapy (Figure 5a). Four years after brachytherapy, the tumour had decreased in thickness, chorioretinal degeneration

surrounded the tumour, optic nerve pallor was present and visual acuity was counting Fingers at 3 feet (Figure 5b). Angiography showed obliteration of small, medium and large blood vessels in the retina and choroid, as well as remodeling of the remaining large vessels (Figure 5c).

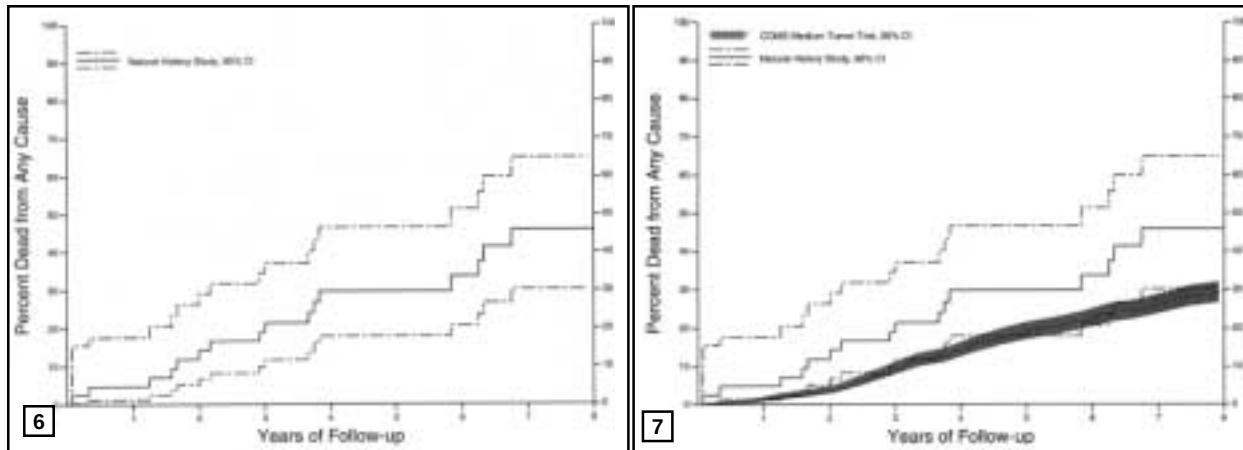


Figure 6. Natural History Study. Cumulative proportion of patients eligible for the I-125 brachytherapy trial who deferred melanoma treatment or had no melanoma treatment and who died by time after determination of eligibility for the COMS brachytherapy trial. Enrolled Natural History Study patients (N=42) with 95% CI; **Figure 7.** Natural History Study and COMS I-125 Brachytherapy Trial. Cumulative proportion of patients with medium choroidal melanoma who died by time after determination of eligibility for the COMS I-125 brachytherapy trial. Enrolled COMS I-125 brachytherapy trial patients (COMS Medium Tumour Trial, N=1,317) and Natural History Study patients who deferred melanoma treatment or had no melanoma treatment (N=42) with 95% CI.

As a further adjunct to the I-125 brachytherapy trial, a natural history study was performed as a prospective non-randomised multicentre cohort study. The natural history study evaluated mortality in patients who were eligible for the I-125 brachytherapy trial but chose to defer treatment or receive no melanoma treatment.¹¹ Forty-two patients with medium-sized choroidal melanoma (42 eyes) chose to defer or receive no melanoma treatment and were followed for a median of 5.3 years (range, 4 – 10.7 years). Twenty-two (52%) of the 42 patients who declined treatment initially received subsequent melanoma treatment and 20 (48%) of the 42 patients had no melanoma treatment.

For the 42 melanoma patients who deferred or had no melanoma treatment, vital status was known at five and ten years for 32 and 20 patients, respectively; the Kaplan-Meier estimate of 5-year all-cause mortality was approximately 30% (95% CI, 18-47%; Figure 6).¹¹ This was greater than the I-125 brachytherapy trial 5-year all-cause mortality of 18% (95% CI, 16-20%), but the difference was not statistically significant (Figure 7). After adjusting for patient age and melanoma longest basal diameter, the 5-year risk of death for the 42 patients who deferred or had no treatment versus the I-125 brachytherapy patients was 1.54 (95% CI, 0.93 to 2.56). In aggregate, the patients who deferred or had no treatment had a 12% greater 5-year mortality and an approximately 50% greater risk of death, which is a probative but not conclusive evidence of a beneficial effect of medium-sized melanoma treatment.¹¹

Overlapping 95% Confidence Intervals may have stemmed from the discrepant sample size. The relatively small sample size of 42 patients who deferred or had no

treatment results in reduced power to detect a statistically significant difference between 18% and 30% (a difference of 12%) was only 44%, but there was an 80% power to detect a difference of 20% or more.¹¹

Taken together, the greater mortality and higher relative risk of death among patients who deferred or had no treatment are evidence that tends to prove the beneficial life-extending effect of melanoma treatment. Although the evidence is probative and not conclusive, it forms a reasonable basis for recommending ocular treatment of medium-sized melanoma. However, the similarity of 5-year outcome emphasises the need for longterm follow-up (≥ 5 years) to determine the efficacy of melanoma treatment by any means.

The rather similar 5-year mortality for treated and not treated medium-sized choroidal melanoma and calculations of tumour doubling time suggest that metastatic dissemination may occur at an early stage of choroidal melanoma.¹²⁻¹⁶ If metastasis has occurred, ocular treatment will control the tumour in the eye but have no effect on the growth of metastatic tumours or longterm survival.

Comparison between 5-year mortality of treated and not treated medium melanoma patients is important in understanding the natural history of choroidal melanoma, but the goal of choroidal melanoma therapy is the maintenance of normal life expectancy. Therefore, based on uncertain time of tumour metastasis and observed differences in 5-year mortality in COMS reports of small sized,¹⁷ medium,^{8,9} and large choroidal melanoma^{15,18} (Table 4), as well as many other studies,^{19,20} every effort should be made to achieve

**Table 4. Five-year mortality in COMS reports of small, medium and large choroidal melanoma**

Tumour size	Study design and treatment	Patients	Five-year all-cause mortality
Small choroidal melanoma ¹⁷ 1.0 – 3.0 mm in apical height and at least 5 mm in basal diameter	Case series discretionary treatment	204	6% (95% CI, 9-20%)
Medium choroidal melanoma ^{8,9} 2.5 – 10.0 mm in apical height and no more than 16 mm in basal diameter	Randomized clinical trial I-125 brachytherapy or enucleation	1,317	18% (95% CI, 16-20%)
Medium choroidal melanoma ¹¹ 2.5-10.0 mm in apical height and no more than 16 mm in basal diameter	Cohort natural history study deferred melanoma treatment or no melanoma treatment	42	30% (95% CI, 18-40%)
Large choroidal melanoma ^{15, 18} > 10.0 mm in apical height or >16 mm in basal diameter	Randomised clinical trial enucleation with or without pre-enucleation radiation	1,003	40% (95% CI, 37-44%)

accurate diagnosis and treatment of choroidal melanoma at small size. In addition, to increase longterm survival, ocular treatment of choroidal melanoma must be combined with measures to detect and treat micrometastasis.¹¹

In summary, this Golden Jubilee Lecture celebrates the Indian Journal of Ophthalmology as the peer reviewed scientific voice of the All India Ophthalmological Society and emphasises the growing importance of evidence-based medicine as the basis for clinical practice. To illustrate evidence-based research, COMS randomised clinical trials and cohort studies were described. The COMS randomised trial of I-125 brachytherapy for medium-sized choroidal melanoma demonstrated a 5-year all-cause mortality of 18% (95% CI, 16-20%) and no significant difference in mortality after treatment with I-125 brachytherapy or enucleation. An adjunctive cohort study of 623 eyes treated with radiation showed that approximately 50% of treated eyes had a substantial loss of visual acuity (six lines or more of acuity or visual

acuity of 6/60 or less) three years after brachytherapy. A cohort natural history study of 42 patients who were eligible for the I-125 brachytherapy trial but chose to defer treatment or receive no treatment, had a 5-year all-cause mortality of 30% (95% CI, 18-47%). The greater mortality and approximately 50% higher 5-year death rate among patients who deferred melanoma or had no choroidal melanoma treatment are probative but not conclusive evidence of a life-extending benefit of treatment for medium choroidal melanoma. To increase long term survival, choroidal melanoma should be diagnosed and treated at an early stage when there is less likelihood of metastasis and ophthalmologists must strive to combine ocular treatment with measures to detect and treat micrometastatic disease.

As the quest for evidence-based medical practice advances, the Indian Journal of Ophthalmology has particular relevance to the populations, cultures and geographic variations of India as well as far reaching relevance to ophthalmic practice worldwide.

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